

Logistic regression interpretation

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Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

Age_i = age (in years)

Random component: $Y_i \sim \text{Bernoulli}(\pi_i)$

Systematic component: $\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 Age_i$

Where did the ε term go?

Linear regression, written two ways:

Option 1:

$$Y_i = \underbrace{\beta_0 + \beta_1 X_i}_{\text{line}} + \underbrace{\varepsilon_i}_{\text{random error}} \quad \varepsilon_i \sim N(0, \sigma^2)$$

random line
(captures variability
in response)

systematic component random component

Option 2:

$$Y_i \sim N(\mu_i, \sigma^2) \quad \leftarrow \text{captures variability in response}$$

random component

$$\mu_i = \beta_0 + \beta_1 X_i \quad \leftarrow \text{captures overall trend}$$

systematic component

when we write a parametric model with

the random & systematic components
(option 2), we don't need to add an " ε " term

Fitting the logistic regression model

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Age}_i$$

"generalized
linear model"



```
m1 <- glm(Dengue ~ Age, data = dengue,  
          family = binomial)  
summary(m1)
```

tells us distribution of response

for logistic regression, use binomial
family

Fitting the logistic regression model

population
model

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Age}_i$$

```
m1 <- glm(Dengue ~ Age, data = dengue,  
           family = binomial)  
summary(m1)
```

...

Coefficients:

##		Estimate	Std. Error	(z) value	Pr(> z)	
##	(Intercept)	-2.454345	0.075068	-32.70	<2e-16	***
##	Age	0.217312	0.008826	24.62	<2e-16	***

z-statistic not
t-statistic

fitted
model

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{Age}_i$$

Activity 1

Work on the activity (handout) with a neighbor, then we will discuss as a class.

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{Age}_i$$

Does the probability of dengue increase or decrease with age?

increase!

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

What are the predicted odds of dengue for a 5 year old patient?

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22(5) = -1.35$$

$$\text{odds} : \frac{\hat{\pi}_i}{1 - \hat{\pi}_i} = e^{-1.35} = 0.26$$

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

For a 5 year old patient: is the predicted *probability* of dengue > 0.5 , < 0.5 , or $= 0.5$?

$$\hat{\pi}_i = 0.5 \quad \Leftrightarrow \quad \text{odds} = 1 \quad \Leftrightarrow \quad \log\text{-odds} = 0$$

$$\hat{\pi}_i < 0.5 \quad \Leftrightarrow \quad \text{odds} < 1 \quad \Leftrightarrow \quad \log\text{-odds} < 0$$

$$\hat{\pi}_i > 0.5 \quad \Leftrightarrow \quad \text{odds} > 1 \quad \Leftrightarrow \quad \log\text{-odds} > 0$$

$$\text{odds} < 1 \quad \Rightarrow \quad \hat{\pi}_i < 0.5$$

Calculating probability

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

$\text{odds} = \frac{\text{prob}}{1 - \text{prob}}$
 $\text{prob} = \frac{\text{odds}}{1 + \text{odds}} = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$

Predicted *probability* of dengue for a 5 year old patient?

$$\text{odds} = \frac{\hat{\pi}_i}{1 - \hat{\pi}_i} = 0.26$$

$$\hat{\pi}_i = 0.26(1 - \hat{\pi}_i)$$

$$\rightarrow \hat{\pi}_i = 0.26 - 0.26 \hat{\pi}_i$$

$$\Rightarrow \hat{\pi}_i + 0.26 \hat{\pi}_i = 0.26$$

$$\Rightarrow \hat{\pi}_i (1 + 0.26) = 0.26$$

$$\Rightarrow \hat{\pi}_i = \frac{0.26}{1 + 0.26} \approx 0.21$$

Activity 2

Work on the activity (handout) with a neighbor, then we will discuss as a class.

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

What is the predicted probability of dengue for a 10 year old patient?

$$\hat{\pi}_i = \frac{\text{odds}}{1 + \text{odds}}$$

$$\text{odds} = 0.78$$

$$\Rightarrow \hat{\pi}_i = \frac{0.78}{1 + 0.78} \approx 0.44$$

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Suppose we want to identify patients for whom the predicted probability of dengue is at least 0.5. What age range should we focus on?

want: log-odds > 0

$$\Rightarrow -2.45 + 0.22 \text{ Age}_i > 0$$

$$\Rightarrow \text{Age} > 11.14$$

Age is a whole # in the data, so
look at patients 12-15 (study only examines patients up to 15)

Activity

increasing age by one year
multiplying odds by a factor of $e^{0.22} = 1.25$

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{Age}_i$$

Compare the odds of dengue for a 12 year old patient to the odds of dengue for an 11 year old patient. What do you notice?

$$\frac{\text{odds when age} = 12}{\text{odds when age} = 11} = 1.25 = \frac{\text{odds when age} = 11}{\text{odds when age} = 10}$$

In general:

$$\frac{\text{odds when age} = x+1}{\text{odds when age} = x} = \frac{e^{-2.45 + 0.22(x+1)}}{e^{-2.45 + 0.22x}} = e^{0.22} = 1.25$$

Interpretation

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Slope:

- + A one-year increase in age is associated with an increase in the *log odds* of dengue of 0.22.
- + A one-year increase in age is associated with an increase in the *odds* of dengue by a *factor* of $e^{0.22} = 1.25$.

Intercept:

Interpretation

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Slope:

- + A one-year increase in age is associated with an increase in the *log odds* of dengue of 0.22.
- + A one-year increase in age is associated with an increase in the *odds* of dengue by a *factor* of $e^{0.22} = 1.25$.

Intercept:

- + when age is 0, the estimated log odds of dengue is -2.45
- + when age is 0, the estimated odds of dengue is $e^{-2.45} = 0.086$

Adding more variables

Now let's add WBC as a variable to the model:

```
m2 <- glm(Dengue ~ Age + WBC, data = dengue,  
           family = binomial)  
summary(m2)
```

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = 0.34 + 0.15 \text{ Age}_i - 0.31 \text{ WBC}_i$$

How should I interpret each coefficient in the fitted model?

0.15 : A one-year increase in age is associated with an increase in odds of dengue by a factor of $e^{0.15}$, holding WBC fixed

Adding more variables

What about a *categorical* predictor?

```
m3 <- glm(Dengue ~ Age + Abdominal, data = dengue,  
          family = binomial)  
summary(m3)
```

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -0.25 + 0.22 \text{ Age}_i + 0.21 \text{ Abdominal}_i$$

How should I interpret the coefficient on abdominal pain?

For next week

Reading: in the textbook, read

+ sections 6.7.1 - 6.7.3

+ sections 2.2 - 2.4

Class on Monday: lab day (most of the class period to work on HW 1)